

Statistical Testing Results and Interpretation

1. Chi-Square Test: Market Segment vs Booking Cancellation

Results

- Chi-Square Statistic: 8497.22
- P-Value: 0.0000
- Degrees of Freedom: 7

Interpretation

The p-value is significantly less than the standard significance level of 0.05. Therefore, the null hypothesis is rejected.

This indicates a statistically significant relationship between Market Segment and Booking Cancellation status.

Business Insight

Different market segments exhibit different cancellation behaviors. Online Travel Agency (OTA) customers may have higher cancellation rates compared to direct or corporate bookings due to more flexible booking policies. Hotels should develop segment-specific pricing and retention strategies to reduce cancellation risk.

2. Chi-Square Test: Customer Type vs Booking Cancellation

Results

- Chi-Square Statistic: 2222.50
- P-Value: 0.0000
- Degrees of Freedom: 3

Interpretation

Since the p-value is less than 0.05, the null hypothesis is rejected.

A statistically significant association exists between Customer Type and Booking Cancellation status.

Business Insight

Transient customers tend to cancel more frequently than contract or group customers. Hotels can improve forecasting accuracy and occupancy management by tailoring booking policies to specific customer types.

3. Independent T-Test: ADR vs Booking Cancellation

Results

- T-Statistic: 16.45
- P-Value: 9.68×10^{-61}

Interpretation

The p-value is extremely small and well below 0.05, indicating a statistically significant difference in Average Daily Rate (ADR) between canceled and non-canceled bookings.

Business Insight

Room pricing has a measurable impact on cancellation behavior. Hotels should continuously monitor pricing strategies and customer sensitivity to room rates when implementing dynamic pricing models.

Overall Conclusion

The statistical analysis confirms that several variables significantly influence hotel booking cancellations.

Key factors include:

1. Market Segment
2. Customer Type
3. Average Daily Rate (ADR)
4. Lead Time (supported through EDA and correlation analysis)
5. Booking Changes (supported through EDA and correlation analysis)

These findings demonstrate that booking cancellations are influenced by both customer characteristics and pricing-related factors. The identified variables should be prioritized in predictive modeling and hotel revenue optimization strategies.